

All numerical answers should have the correct number of significant figures

Points: ____/15

1. Humans have unwittingly used uranium since antiquity. Yellow glass, 1.05% by mass uranium(V) oxide (U_2O_5), was discovered in a Roman villa dating to the 1st century A.D. If the glass weighs 30.6 g, how many grams of uranium are present and could be extracted from the glass? Assume that this extraction proceeds with perfect efficiency. Show your work (**4 pts**) $\text{U} = 238.03 \text{ amu}$
 $\text{O} = 16.00 \text{ amu}$

2. Brilliant Blue FCF ($\text{MM} = 792.85 \text{ g/mol}$) is the common name for a molecular compound that is used in blue food coloring.
 - a. i. and ii. are mixed resulting in iii. Identify each of the three following as solute, solvent or solution. (**3 pts**)
 - i. 101 mg Brilliant Blue FCF
 - ii. 101 mL Water (H_2O)
 - iii. Mixture composed of 101 mg Brilliant Blue FCF dissolved in 101 mL H_2O
 - b. What is the molarity (mol/L) of Brilliant Blue FCF in the mixture in iii? Assume that the total volume of solution is 101 mL. (**2 pts**) $1 \text{ mg} = 10^{-3} \text{ g}$ $1 \text{ mL} = 10^{-3} \text{ L}$

3. Liquid naphthalene (C_{10}H_8) reacts (COMBUSTS) with gaseous oxygen (O_2) to form carbon dioxide gas and liquid water. Write a balanced chemical equation for this reaction and include the phases of matter. CHECK YOUR WORK (**3 pts**)

4. Consider the reaction $2\text{Sb(s)} + 3\text{I}_2\text{(s)} \rightarrow 2\text{SbI}_3\text{(s)}$
 - a. What is the limiting reactant when 2.40 g of I_2 and 1.20 g of Sb are mixed? Show your work (**2 pts**)

 - b. What is the theoretical yield of SbI_3 in part a? (**1 pt**)